

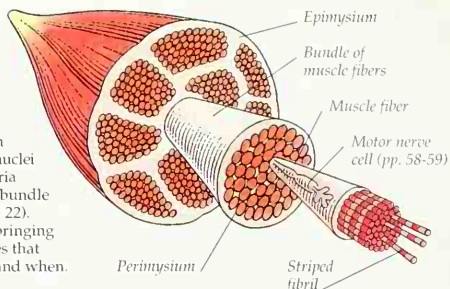


MR. MUSCLE

Bodybuilders take pride in developing, emphasizing, and flexing their muscles. Many of the body's superficial muscles can be identified, beautifully defined, rippling just below the skin. When a muscle is tensed and shortened, its main body, or belly, bulges and becomes wider.

INSIDE A MUSCLE

Each skeletal muscle is wrapped in a tough sheath – the epimysium. Within are bundles of hairlike muscle fibers, each wrapped in its own sheath, the perimysium. The fibers are giant multi-cells, some more than 0.4 in (10 mm) long, containing several nuclei and huge numbers of mitochondria (pp. 12-13). In turn, one fiber is a bundle of fibrils, the contracting units (p. 22). Muscle has a rich blood supply, bringing energy for contraction, and nerves that instruct which fibers to contract and when.



Trapezius muscle is attached to the skull at one end

Trapezius braces the shoulders and pulls back the head

Infraspinatus pulls arm against body

Gluteus maximus straightens the hip in walking and running

Gastrocnemius lifts the heel and bends the knee

Calcaneus tendon is also called Achilles tendon

Superficial muscles, rear view

Triceps brachii straightens elbow

Diaphragm (p. 24)

Gluteus medius pulls thigh out to the side

Adductor magnus pulls the leg toward the center of the body

Deep muscles, front view

DEEP MUSCLES

If the superficial and middle muscles are peeled away, the deepest muscles are exposed. These lie directly next to the bones they pull, and their points of anchorage on the bones are clearly seen. Some are flat and sheet-shaped; others have the classic bulging shape.

Semispinalis capitis tilts the head to look up

Intercostal muscles move ribs in breathing (p. 25)

Vastus lateralis bends and stabilizes knee

Tibialis posterior counteracts sway when standing on one foot

Calf muscle tendons pass through ankle into foot

Deep muscles, rear view

Flexor hallucis curls sole and toes downward